

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-15

Faculty

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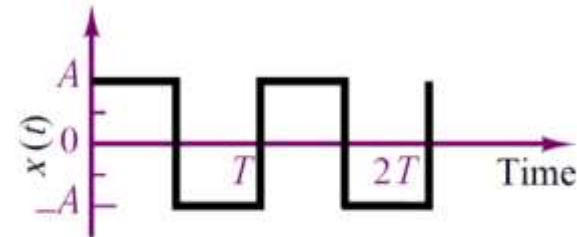
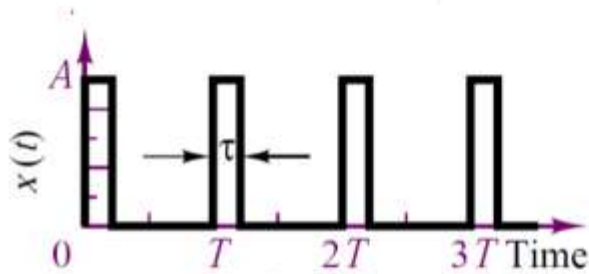
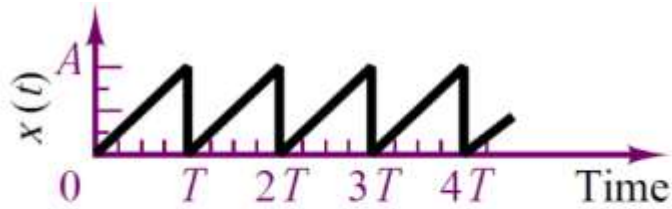
Associate Professor G-1

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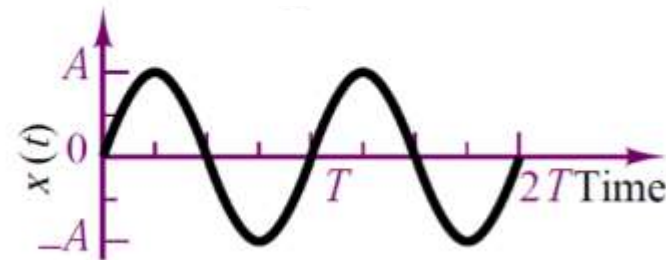
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Some Typical signals

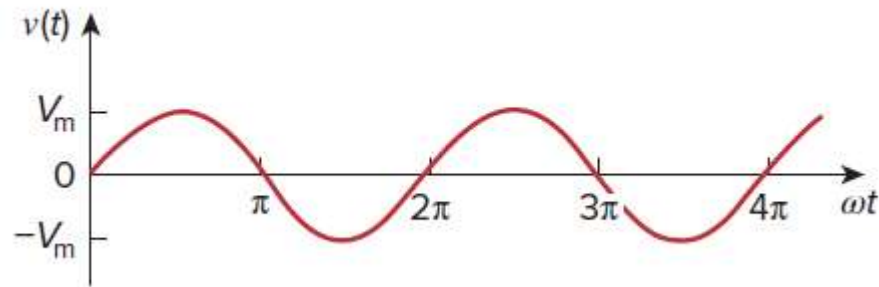


- These signals appear frequently in practical applications and are a useful approximation of many physical phenomena.
- **Cycle and symmetrical waveform**
- **Period and Frequency**



Average Values

- It is the equivalent steady state value of time varying waveform which could transfer the same amount of charge as given by the original waveform for the definite period of time.



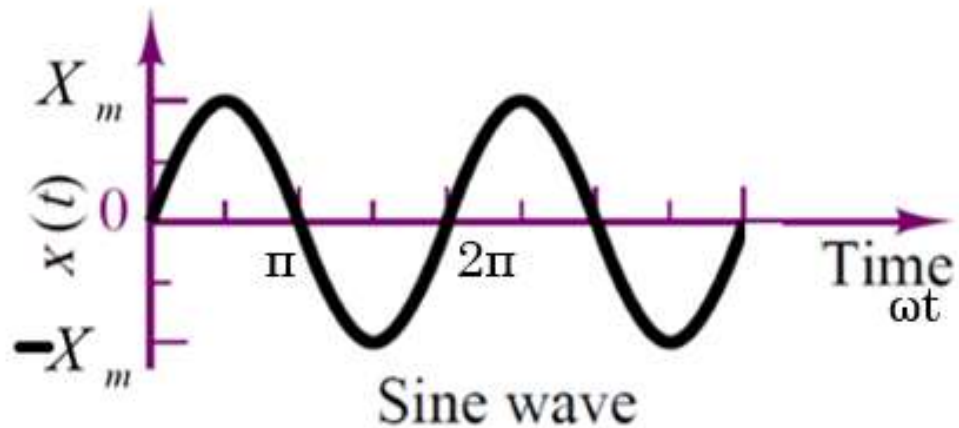
I.Average Values

- The average value is the ratio between area under the curve and the cycle period.
- If the cycle is symmetrical with respect to time axis, then we should consider only half cycle period. Considering full cycle will lead to the trivial mean of zero.
- The time averaged value of a signal $v(t)$ is:

$$V_{av} = \frac{1}{T} \int_0^T v(t) dt$$

1. Find the Average value of Sinusoidal Wave

1. Find the Average value of Sinusoidal Wave(Half Cycle)



$$x(\omega t) = X_m \sin \omega t$$

$$X_{av} = \frac{1}{\pi} \int_0^{\pi} x(\omega t) d\omega t$$

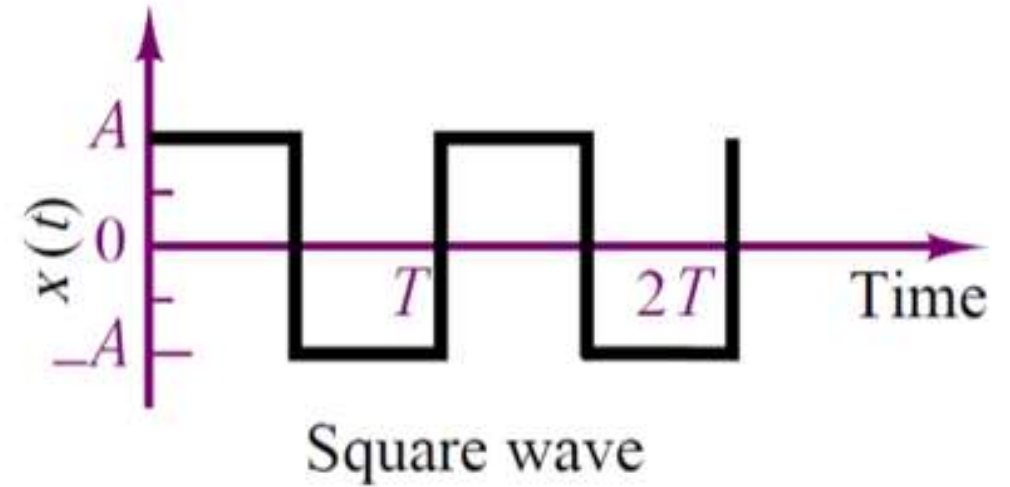
$$X_{av} = \frac{1}{\pi} \int_0^{\pi} X_m \sin \omega t d\omega t$$

$$X_{av} = \frac{X_m}{\pi} [-\cos \omega t]_0^{\pi}$$

$$= \frac{2X_m}{\pi}$$

2. Find the Average value of Square Wave

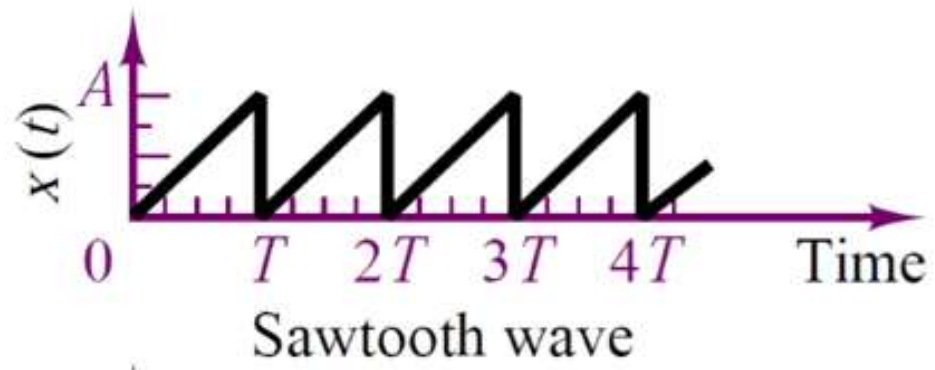
2. Square wave(Half Cycle)



$$X_{av} = \frac{A \times T/2}{T/2} = A$$

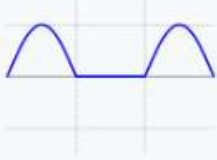
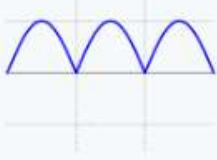
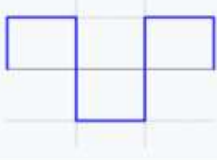
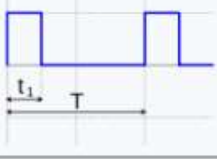
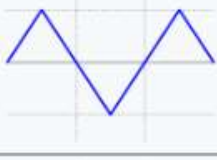
3. Find the Average value of Sawtooth Wave

3. Sawtooth Wave:



$$X_{av} = \frac{\left(\frac{1}{2} \times A \times T \right)}{T} = \frac{A}{2}$$

Find the average value of certain periodic signals

Waveform	Image	ARV
Sine wave		$a \frac{2}{\pi} [2]$
Half-wave rectified sine		$\frac{a}{\pi}$
Full-wave rectified sine		$a \frac{2}{\pi}$
Square wave, constant value		a
Pulse wave		aD
Triangle wave		$\frac{a}{2}$
Sawtooth wave		$\frac{a}{2}$